```
/*////////////////////////////////////////////////////////////////////////////
//////////////////
//
// Project:    KSvision
//
// Module:     KSerror.h
//
// Comments:   Error constants for KSvision.
//
// Copyright:  Kasalis Inc.
//
////////////////////////////////////////////////////////////////////////////
//////////////////*/
#pragma once

#define ERR_WAITING                          1 // KSVision returns this error to the PIXID system, PIXID needs to retry because of waiting for outside site

#define ERR_NONE                             0

// configuration errors
#define ERR_CONFIG_NOT_LOADED               -1
#define ERR_CONFIG_EXISTS_KSINI             -2
#define ERR_CONFIG_EXISTS_KSGLOBAL          -3
#define ERR_CONFIG_EXISTS_KSFIXTURE         -4
#define ERR_CONFIG_EXISTS_KSPROCESS         -5
#define ERR_CONFIG_EXISTS_KSDISPENSE        -6
#define ERR_CONFIG_LOAD_EDIT_FILE           -7
#define ERR_CONFIG_SAVE_EDIT_FILE           -8
#define ERR_CONFIG_LOAD_EDIT_TEXT           -9
#define ERR_CONFIG_SAVE_EDIT_TEXT           -10
#define ERR_CONFIG_LOAD_TEXT_TOO_LARGE      -11
#define ERR_CONFIG_SAVE_TEXT_TOO_LARGE      -12
#define ERR_CONFIG_MEM_ALLOC                -13
#define ERR_CONFIG_PARAM_NOT_FOUND          -14
#define ERR_CONFIG_EXISTS_CONFIG_FOLDER     -15
#define ERR_CONFIG_EXISTS_DATA_FOLDER       -16
#define ERR_CONFIG_EXISTS_IMAGE_FOLDER      -17
#define ERR_CONFIG_EXISTS_BRVCLOADER        -18
#define ERR_CONFIG_KSDISPENSE_OPEN          -19
#define ERR_CONFIG_KSDISPENSE_GET_SIZE      -20
#define ERR_CONFIG_KSDISPENSE_PARSE         -21
#define ERR_CONFIG_KSFIXTURE_OPEN           -22
#define ERR_CONFIG_KSFIXTURE_GET_SIZE       -23
#define ERR_CONFIG_KSFIXTURE_PARSE          -24
#define ERR_CONFIG_KSFIXTURE_DUP_ENTRIES    -25
#define ERR_CONFIG_MACH_BAD_MODEL_TYPE      -26
#define ERR_CONFIG_PARAM_NOT_VALID          -27
#define ERR_CONFIG_EXISTS_CTR_INI           -28
#define ERR_CONFIG_EXISTS_CTR_CAM_FILE_PARAM -29
#define ERR_CONFIG_DUAL_HEAD_DUAL_FIXTURE   -30
```

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 49  #define ERR_CONFIG_EXT_LIB_DUAL_FIXTURE      -31
 50  #define ERR_CONFIG_INTERNAL_ROBOT_MES_ROBOT  -32
 51  #define ERR_CONFIG_DOUBLET_ALIGN_ARRAY_GRIP  -33
 52  #define ERR_CONFIG_KSDISPENSE_PAT_UNDEFINED  -34
 53  #define ERR_CONFIG_KSDISPENSE_PRE_UNDEFINED  -35
 54  #define ERR_CONFIG_KSDISPENSE_PAT_DUPLICATE  -36
 55  #define ERR_CONFIG_KSDISPENSE_PRE_DUPLICATE  -37
 56  #define ERR_CONFIG_KSDISPENSE_PAT_OUTRANGE   -38
 57  #define ERR_CONFIG_KSDISPENSE_PRE_OUTRANGE   -39
 58  #define ERR_CONFIG_INVALID_SCANNER_CONFIG    -40
 59  #define ERR_CONFIG_UNDEFINED_DATAREC_TYPE    -41
 60  #define ERR_CONFIG_INVALID_GREYSCALE_FACTORS -42
 61  #define ERR_CONFIG_PARAM_MISSING_VALUES      -43
 62  #define ERR_CONFIG_PROC_TRACK_NOT_SUPPORTED  -44
 63  #define ERR_CONFIG_KSDISPENSE_FX_OUTRANGE    -45
 64  #define ERR_CONFIG_2ND_DISPLAY_LAUNCH_ERROR  -46
 65  #define ERR_CONFIG_2ND_DISPLAY_IMG_SIZE_ERR  -47
 66  #define ERR_CONFIG_EXISTS_UPLOAD_FOLDER      -48
 67  #define ERR_CONFIG_PARAM_TOO_MANY_VALUES     -49
 68  // Additional configuration error group starts from -850
 69
 70  // user table errors
 71  #define ERR_USER_TABLE_EXISTS                -50
 72  #define ERR_USER_TABLE_OPEN                  -51
 73  #define ERR_USER_TABLE_GET_SIZE              -52
 74  #define ERR_USER_TABLE_MEM_ALLOC             -53
 75  #define ERR_USER_WRONG_NAME_LENGTH           -54
 76  #define ERR_USER_WRONG_PWD_LENGTH            -55
 77  #define ERR_USER_WRONG_ACCESS_LEVEL          -56
 78  #define ERR_USER_ENGINEER_NOT_FOUND          -57
 79  #define ERR_USER_TECHNICIAN_NOT_FOUND        -58
 80
 81  // device camera errors
 82  #define ERR_CAM_DUT_INIT_EXISTS_XSDAT        -101
 83  #define ERR_CAM_DUT_INIT_EXISTS_INI          -102
 84  #define ERR_CAM_DUT_INIT_OPEN_INI            -103
 85  #define ERR_CAM_DUT_INIT_GET_BUFFER_SIZE     -104
 86  #define ERR_CAM_DUT_INIT_MEM_ALLOC           -105
 87  #define ERR_CAM_DUT_INIT_BAD_CAM_TYPE        -106
 88  #define ERR_CAM_DUT_INIT_BAD_IMG_FMT         -107
 89  #define ERR_CAM_DUT_INIT_WRONG_IMG_FMT       -108
 90  #define ERR_CAM_DUT_INIT_WRONG_PIX_BITS      -109
 91  #define ERR_CAM_DUT_INIT_BAD_FRAME_SIZE      -110
 92  #define ERR_CAM_DUT_INIT_NOT_OPEN            -111
 93  #define ERR_CAM_DUT_INIT_EXCEPTION           -112
 94  #define ERR_CAM_DUT_INIT_NO_CAMS_FOUND       -113
 95  #define ERR_CAM_DUT_INIT_NO_VIDEO_SIGNAL     -114
 96  #define ERR_CAM_DUT_INIT_BAD_HDW_ID          -115
 97  #define ERR_CAM_DUT_INIT_BAD_INI_IMG_FMT     -116
 98  #define ERR_CAM_DUT_INIT_LOAD_REGS           -117
 99  #define ERR_CAM_DUT_INIT_DEVICE_FUNCTION     -118
100  #define ERR_CAM_DUT_INIT_INI_IMG_FMT_NA      -119
```

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101  #define ERR_CAM_DUT_INIT_BAD_INI_RCC_FMT     -120
102  #define ERR_CAM_DUT_INIT_NO_MIU_FOUND        -121
103  #define ERR_CAM_DUT_INIT_DUPL_MIU_FOUND      -122
104  #define ERR_CAM_DUT_INIT_FIND_MIUS           -123
105  #define ERR_CAM_DUT_INIT_MIU_OPEN_ALIGN      -124
106  #define ERR_CAM_DUT_INIT_MIU_OPEN_DISP       -125
107  #define ERR_CAM_DUT_INIT_MIU_ALIGN_NF        -126
108  #define ERR_CAM_DUT_INIT_MIU_DISP_NF         -127
109  #define ERR_CAM_DUT_INIT_RAW_BUF_EMPTY       -128
110  #define ERR_CAM_DUT_INIT_RAW_BAD_PARAMS      -129
111  #define ERR_CAM_DUT_INIT_RAW_BAD_IMG_TYPE    -130
112  #define ERR_CAM_DUT_INIT_RAW_BAD_CAM_TYPE    -131
113  #define ERR_CAM_DUT_INIT_WRONG_IMG_SIZE      -132
114  #define ERR_CAM_DUT_INIT_AARDVARK_NA         -133
115  #define ERR_CAM_DUT_INIT_INI_IICFPID_NA      -134
116  #define ERR_CAM_DUT_INIT_INI_MIPIFPVAL_NA    -135
117  #define ERR_CAM_DUT_INIT_LOAD_3RD_PARTY_LIB  -136
118  #define ERR_CAM_DUT_INIT_GET_3RD_PARTY_FNC   -137
119  #define ERR_CAM_DUT_INIT_READ_HDW_ID         -138
120  #define ERR_CAM_DUT_INIT_DEVICE_TEMPERATURE  -139
121  #define ERR_CAM_DUT_INIT_MIU_SPARE_NF        -140
122  #define ERR_CAM_DUT_INIT_MIU_OPEN_SPARE      -141
123  #define ERR_CAM_DUT_INIT_MIU_OPEN            -142
124  #define ERR_CAM_DUT_INIT_CROP_DATA_OUTOF_RANGE     -143
125  #define ERR_CAM_DUT_INIT_INVALID_CROP_SIZE_10BIT   -144
126  #define ERR_CAM_DUT_INIT_INVALID_CROP_SIZE_12BIT   -145
127  #define ERR_CAM_DUT_INIT_IMG_PROCESS         -146
128
129  // machine vision camera errors
130  #define ERR_CAM_VIS_INIT_CREATE_CONTEXT      -201
131  #define ERR_CAM_VIS_INIT_NUM_CAMS_GET        -202
132  #define ERR_CAM_VIS_INIT_NUM_CAMS_ZERO       -203
133  #define ERR_CAM_VIS_INIT_SERNUM_NF           -204
134  #define ERR_CAM_VIS_INIT_CONNECT             -205
135  #define ERR_CAM_VIS_INIT_CAM_INFO            -206
136  #define ERR_CAM_VIS_INIT_CREATE_IMG_BUF      -207
137  #define ERR_CAM_VIS_INIT_START_CAPTURE       -208
138  #define ERR_CAM_VIS_INIT_GET_FRAME           -209
139  #define ERR_CAM_VIS_INIT_SIZE_INVALID        -210
140  #define ERR_CAM_VIS_INIT_FORMAT_INVALID      -211
141  #define ERR_CAM_VIS_INIT_NOT_OPEN            -212
142  #define ERR_CAM_VIS_INIT_DUPLICATE_SN        -213
143  #define ERR_CAM_VIS_INIT_GET_PROP_INFO       -214
144  #define ERR_CAM_VIS_INIT_SET_PROP            -215
145  #define ERR_CAM_VIS_INIT_PROP_NOT_PRESENT    -216
146  #define ERR_CAM_VIS_SET_EXPOSURE             -217
147  #define ERR_CAM_VIS_INIT_EMBEDDED_INFO       -218
148  #define ERR_CAM_VIS_INIT_MEM_ALLOC           -219
149  #define ERR_CAM_VIS_INIT_COMPOSITE_CAM_SN_FORMAT -220
150  #define ERR_CAM_VIS_INIT_COMPOSITE_CAM_NOT_OPEN  -221
151  #define ERR_CAM_VIS_INIT_SET_GRAB_TIMEOUT        -222
152  #define ERR_CAM_VIS_INIT_GET_EXPOSURE        -223
```

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153 #define ERR_CAM_VIS_INVALID_CAM_TYPE         -224
154 #define ERR_CAM_VIS_SET_GAIN                 -225
155 #define ERR_CAM_VIS_GET_GAIN                 -226
156 #define ERR_CAM_VIS_SET_OFFSET               -227
157 #define ERR_CAM_VIS_GET_AUTOINFO             -228
158 #define ERR_CAM_VIS_PIXEL_CLOCK              -229
159 #define ERR_CAM_VIS_INIT_COLOR_MODE          -230
160 #define ERR_CAM_VIS_INIT_ACTIVE_MEM          -231
161 #define ERR_CAM_VIS_INIT_SOFTTRIG            -232
162 #define ERR_CAM_VIS_INIT_IDS_SERNUM_NF       -233
163 #define ERR_CAM_VIS_SET_TIMEOUT              -234
164 #define ERR_CAM_VIS_SET_TRIGGER_DELAY        -235
165 #define ERR_CAM_VIS_SET_EXTERNAL_TRIGGER     -236
166 #define ERR_CAM_VIS_INIT_EVENT               -237
167 #define ERR_CAM_VIS_ENABLE_EVENT             -238
168 #define ERR_CAM_VIS_UNSUPPORTED_COLOR_MODE   -239
169
170 // image processing errors
171 #define ERR_IMAGE_LOAD_FAIL                  -301
172 #define ERR_IMAGE_SAVE_FAIL                  -302
173 #define ERR_IMAGE_LOAD_INVALID_FILE_NAME     -303
174 #define ERR_IMAGE_SAVE_INVALID_FILE_NAME     -304
175 #define ERR_IMAGE_GRAB_FAILED_DUT            -305
176 #define ERR_IMAGE_GRAB_FAILED_VIS            -306
177 #define ERR_IMAGE_GRAB_BAD_ARRAY_SIZE        -307
178 #define ERR_IMAGE_FRAME_NUM_INVALID          -308
179 #define ERR_IMAGE_FRAME_NO_EMPTY_FRAMES      -309
180 #define ERR_IMAGE_FRAME_BUFFER_EMPTY         -310
181 #define ERR_IMAGE_BAD_VIS_CAM_NUM            -311
182 #define ERR_IMAGE_BAD_DUT_CAM_NUM            -312
183 #define ERR_IMAGE_MULT_DUT_CAMS_NA           -313
184 #define ERR_IMAGE_INVALID_BIT_DEPTH          -314
185 #define ERR_IMAGE_UNSUPPORTED_COLOR_MODE     -315
186 #define ERR_IMAGE_NO_IMAGE_DATA              -316
187
188 // general errors
189 #define ERR_CREATE_CLIENT_WINDOW             -401
190 #define ERR_LOAD_KS_CAM_LIBRARY              -402
191 #define ERR_GET_KS_CAM_LIB_FNC_PTR           -403
192 #define ERR_INVALID_TOOL_INDEX               -404
193 #define ERR_FOCUS_NO_FRAMES_TO_ANALYZE       -405
194 #define ERR_LOAD_KS_MACHVIS_LIBRARY          -406
195 #define ERR_GET_KS_MACHVIS_LIB_FNC_PTR       -407
196 #define ERR_LOAD_KS_HEXSIGHT_LIBRARY         -408
197 #define ERR_GET_KS_HEXSIGHT_LIB_FNC_PTR      -409
198 #define ERR_INVALID_ALIGN_MODE               -410
199 #define ERR_INVALID_CALIB_MODE               -411
200 #define ERR_INVALID_DISP_CAL_STEP_NUM        -412
201 #define ERR_KS_HEXISGHT_LIB_NOT_INIT         -413
202 #define ERR_NA_64BIT                         -414
203 #define ERR_INALID_IMAGE_ARRAY_PARAMS        -415
204 #define ERR_PATTERN_NO_MATCHES               -416
```

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205  #define ERR_FUNCTION_NOT_SUPPORTED           -417
206  #define ERR_CAM_DUT_SET_REG_EXISTS_INI       -418
207  #define ERR_CAM_DUT_SET_REG_OPEN_INI         -419
208  #define ERR_CAM_DUT_SET_REG_ARG_INVALID      -420
209  #define ERR_CAM_DUT_SET_REG_ERROR            -421
210  #define ERR_CAM_DUT_SET_REG_MALLOC           -422
211  #define ERR_MAX_WORK_ID                      -423
212  #define ERR_INVALID_ROTARY_TABLE_POS         -424
213  #define ERR_INVALID_FIXTURE_TYPE             -425
214  #define ERR_TEST_NOT_ENABLED                 -426
215  #define ERR_IMAGE_FRAMES_NOT_SAME_SIZE       -427
216  #define ERR_CROP_BORDER_TOO_LARGE            -428
217  #define ERR_GET_KS_I2C_LIB_FNC_PTR           -429
218  #define ERR_INVALID_FOCAL_PLANE              -430
219  #define ERR_CREATE_IMAGE_PROCESS_THREAD      -431
220  #define ERR_UVMEASURE_CREATE_CSV_FILE        -432
221  #define ERR_DATAREC_INVALID_STATION_NO       -433
222  #define ERR_DATAREC_CREATE_CSV_FILE          -434
223  #define ERR_DISPINSPECT_CREATE_IMAGE_FILE    -435
224  #define ERR_ALIGN_FAIL_CALC_ONAXIS_PEAK      -436
225  #define ERR_ALIGN_FAIL_MAX_SCANS_REACHED     -437
226  #define ERR_ALIGN_PEAK_TZ_SCAN_MAX_STEPS     -438
227  #define ERR_EXTLIB_LOAD                      -439
228  #define ERR_EXTLIB_GET_FUNC_PTR              -440
229  #define ERR_EXTLIB_CREATE_IMAGE_FILE         -441
230  #define ERR_EXTLIB_FUNC_DISPENSEON           -442
231  #define ERR_EXTLIB_FUNC_DISPENSEOFF          -443
232  #define ERR_EXTLIB_FUNC_ALIGNMENTON          -444
233  #define ERR_EXTLIB_FUNC_ALIGNMENTOFF         -445
234  #define ERR_EXTLIB_FUNC_INSPECT              -446
235  #define ERR_EXTLIB_FUNC_PARTDONE             -447
236  #define ERR_EXTLIB_FUNC_INDEXREQUEST         -448
237  #define ERR_SCANNERLIB_LOAD                  -449
238  #define ERR_SCANNERLIB_GET_FUNC_PTR          -450
239  #define ERR_SCANNERLIB_WRONG_FIXTURE_POS1    -451
240  #define ERR_SCANNERLIB_WRONG_FIXTURE_POS2    -452
241  #define ERR_SCANNERLIB_WRONG_FIXTURE_POS3    -453
242  #define ERR_SCANNERLIB_WRONG_FIXTURE_POS4    -454
243  #define ERR_LASERREADER_LIB_LOAD             -455
244  #define ERR_LASERREADER_GET_FUNC_PTR         -456
245  #define ERR_LIGHTWAVE_LIB_LOAD               -457
246  #define ERR_LIGHTWAVE_GET_FUNC_PTR           -458
247  #define ERR_LASERSCAN_INVALID_COMMAND        -459
248  #define ERR_LASERSCAN_CREATE_IMAGE_FILE      -460
249  #define ERR_LOAD_SIMULATION_IMAGE            -461
250  #define ERR_DISPENSEWEIGH_LOAD               -462
251  #define ERR_DISPENSEWEIGH_GET_FUNC_PTR       -463
252  #define ERR_DISPENSEWEIGH_UNEXPECTED_RESP    -464
253  #define ERR_DISPENSEWEIGH_WRONG_UNIT         -465
254  #define ERR_DISPENSEWEIGH_STABLE_WGT_TIMEOUT -466
255  #define ERR_DISPENSEWEIGH_CREATE_CSV_FILE    -467
256  #define ERR_KS_HEXISGHT_FILE_NOT_FOUND       -468
```

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257 #define ERR_DOUBLET_FINAL_SCAN_THRESH        -469
258 #define ERR_CAM_DUT_GET_REG_ARG_INVALID      -470
259 #define ERR_CAM_DUT_GET_REG_ERROR            -471
260 #define ERR_MACHINE_LOG_FILE_OPEN            -472
261 #define ERR_DEBUG_LOG_FILE_OPEN              -473
262 #define ERR_MACHINE_LOG_FILE_WRITE           -474
263 #define ERR_DEBUG_LOG_FILE_WRITE             -475
264 #define ERR_KSDATARAY_LIB_LOAD               -476
265 #define ERR_KSDATARAY_GET_FUNC_PTR           -477
266 #define ERR_GET_MACHINE_HDW_ID               -478
267 #define ERR_CREATE_PIXID_SYSTEM_FOLDER       -479
268 #define ERR_ROBOT_LOG_FILE_OPEN              -480
269 #define ERR_ROBOT_LOG_FILE_WRITE             -481
270 #define ERR_INVALID_FIXTURE_NUMBER           -482
271 #define ERR_DUT_SER_NUM_LOG_FILE_OPEN        -483
272 #define ERR_BNC_LIB_LOAD                     -484
273 #define ERR_BNC_LIB_FNC_PTR                  -485
274 #define ERR_TEMPLATE_MATCH_LOAD_IMAGE        -486
275 #define ERR_CREATE_IMAGE_ARCHIVE             -487
276 #define ERR_CREATE_PIXID_UPLOAD_FOLDER       -488
277 #define ERR_ALIGN_SCAN_LOG_FILE_WRITE        -489
278 #define ERR_INVALID_ASSEMBLY_NUMBER          -490
279 #define ERR_ALIGN_OFF_AXIS_WEIGHT_POLYORDER  -491
280 #define ERR_ALIGN_OFF_AXIS_WEIGHT_MODE2_MTF_HI  -492
281 #define ERR_ALIGN_OFF_AXIS_WEIGHT_MODE2_MTF_LO  -493
282 #define ERR_CONFIGBACKUP_CREATE_ZIP_FOLDER          -494
283 #define ERR_CONFIGBACKUP_CREATE_PIXID_LOG_FOLDER    -495
284 #define ERR_CONFIGBACKUP_CREATE_OPEN_FILE_RECORD    -496
285 #define ERR_CONFIGBACKUP_CREATE_OPEN_FILELIST       -497
286 #define ERR_CREATE_PIXIDSYS_FILE                    -498
287 #define ERR_INCR_WORK_ID                            -499
288 #define ERR_INVALID_CALIB_MEM_ALLOC                 -500
289
290 // visualization errors
291 #define ERR_VISU_INVALID_TCP_COMMAND         -501
292 #define ERR_VISU_GET_WINDOW_HANDLE           -502
293 #define ERR_VISU_GET_PROCESS_ID              -503
294 #define ERR_VISU_GET_PROCESS_ACCESS          -504
295 #define ERR_VISU_CLOSE_PROCESS               -505
296 #define ERR_VISU_INVALID_VIEW_MODE           -506
297 #define ERR_VISU_DATA_SIZE_MISMATCH          -507
298
299 // MES interface errors
300 #define ERR_MES_CREATE_PROCESS               -510
301 #define ERR_MES_CONNECT_TO_KSMESCOMM         -511
302 #define ERR_MES_SEND_BUF_TOO_SMALL           -512
303 #define ERR_MES_SEND_FAILED                  -513
304 #define ERR_MES_SEND_RESPONSE_FAILED         -514
305 #define ERR_MES_NOT_CONNECTED                -516
306 #define ERR_MES_CONNECT_AS_HOST              -517
307 #define ERR_MES_WRONG_VERSION                -518
308 #define ERR_MES_DUPLEX_REQUEST_ACK_FAILED    -519
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309 #define ERR_MES_DUPLEX_REQUEST_RESP_FAILED   -520
310 #define ERR_MES_DUPLEX_MES_MODE              -521
311 #define ERR_MES_DUPLEX_EXISTS_KSGOLDSAMPLE   -522
312 #define ERR_MES_DUPLEX_DEVICESTATE_FAIL      -523
313
314 // Python scripting errors
315 #define ERR_ALIGN_PYTHON_FILE_NOT_FOUND      -550
316 #define ERR_ALIGN_PYTHON_EXCEPTION           -551
317 #define ERR_ALIGN_PYTHON_ERROR_CODE          -552
318 #define ERR_ALIGN_PYTHON_FIXTURE_OFFSET_BUF  -553
319
320 // blob errors
321 #define ERR_BLOB_MEM_ALLOC                   -601
322 #define ERR_BLOB_INVALID_LABEL_NUM           -602
323 #define ERR_BLOB_CALIBTOOL_NO_BLOBS          -603
324 #define ERR_BLOB_HOT_SPOT_NOT_FOUND          -604
325 #define ERR_BLOB_HOT_SPOT_TOO_BRIGHT         -605
326
327 // passive/active align errors
328 #define ERR_ALIGN_FAIL_MOTION_LIMIT          -607
329 #define ERR_ALIGN_FAIL_LASER_DIODE_OFF       -608
330
331 // centration errors - object location based
332 #define ERR_CENTRATION_FAIL_FIND_FEATURE     -610
333 #define ERR_CENTRATION_FAIL_LINE_FIT         -611
334 #define ERR_CENTRATION_FAIL_ANGLE_FEATURES   -612
335 #define ERR_CENTRATION_FAIL_MOTION_LIMIT     -613
336 #define ERR_CENTRATION_FAIL_TOUCHING_BORDER  -614
337
338 // centration errors - gradient location based
339 #define ERR_CENTRATION_FAIL_FIND_CENTER      -615
340 #define ERR_CENTRATION_FAIL_TOO_BRIGHT       -616
341 #define ERR_CENTRATION_FAIL_TOO_DARK         -617
342 #define ERR_CENTRATION_SATURATED             -618
343
344 // MTF errors
345 #define ERR_MTF_FEATURE_NOT_FOUND            -620
346 #define ERR_MTF_FEATURE_NOT_VALID            -621
347 #define ERR_MTF_MEM_ALLOC                    -622
348 #define ERR_MTF_BAD_GRADIENT                 -623
349 #define ERR_MTF_EDGE_TOO_CLOSE               -624
350 #define ERR_MTF_FEATURE_TOO_SMALL            -625
351 #define ERR_MTF_REGION_OUTSIDE_FRAME         -626
352 #define ERR_MTF_ANGLE_TOO_SMALL              -627
353 #define ERR_MTF_ANGLE_TOO_LARGE              -628
354 #define ERR_MTF_CANT_INTERPOLATE             -629
355 #define ERR_MTF_REGION_OUTSIDE_ROI           -630
356 #define ERR_MTF_FEATURE_BLOB_NOT_FOUND       -631
357 #define ERR_MTF_FEATURE_CHARACTERISTICS      -632
358 #define ERR_MTF_FEATURE_ANGLE_THRESH         -633
359 #define ERR_MTF_FEATURE_GRADIENT_THRESH      -634
360 #define ERR_MTF_FEATURE_ROTATION             -635
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361 #define ERR_MTF_ROI_NOT_DEFINED              -636
362 #define ERR_MTF_FEATURE_EDGE_LENGTH          -637
363
364 // sensor test general errors
365 #define ERR_SENSORTEST_INVALID_MODE          -641
366
367 // user test errors
368 #define ERR_USERTEST_CREATE_CSV_FILE         -645
369 #define ERR_USERTEST_FAILED                  -646
370
371 // repeatability and stability test errors
372 #define ERR_STABILITY_TEST_CREATE_CSV        -649
373 #define ERR_REPEATABILITY_TEST_CREATE_CSV    -650
374
375 // particle test errors
376 #define ERR_PARTICLETEST_CREATE_CSV_FILE     -651
377 #define ERR_PARTICLETEST_CREATE_IMAGE_FILE   -652
378 #define ERR_PARTICLETEST_TEST_FAILED         -653
379 #define ERR_PARTICLETEST_BAD_STEP_NUM        -654
380 #define ERR_PARTICLETEST_BAD_LUMINANCE_RANGE -655
381
382 // defective pixel test errors
383 #define ERR_PIXELTEST_CREATE_CSV_FILE        -661
384 #define ERR_PIXELTEST_CREATE_IMAGE_FILE      -662
385 #define ERR_PIXELTEST_BAD_CLUSTER_PARAM      -663
386 #define ERR_PIXELTEST_TEST_FAILED            -664
387
388 // dark signal test errors
389 #define ERR_DARKTEST_CREATE_CSV_FILE         -671
390 #define ERR_DARKTEST_CREATE_IMAGE_FILE       -672
391 #define ERR_DARKTEST_TEST_FAILED             -673
392
393 // temporal noise test error
394 #define ERR_NOISETEST_CREATE_CSV_FILE        -681
395 #define ERR_NOISETEST_CREATE_IMAGE_FILE      -682
396 #define ERR_NOISETEST_TEST_FAILED            -683
397
398 // post dispense inspect errors
399 #define ERR_DISPENSETEST_CREATE_CSV_FILE     -691
400 #define ERR_DISPENSETEST_CREATE_IMAGE_FILE   -692
401 #define ERR_DISPENSETEST_TEST_FAILED_BLOB    -693
402 #define ERR_DISPENSETEST_BAD_STEP_NUM        -694
403 #define ERR_DISPENSETEST_TEST_FAILED_CS      -695
404 #define ERR_DISPENSETEST_REMOTE_BAD_DISP     -696
405 #define ERR_DISPENSETEST_REMOTE_FAILED_IMAGE -697
406 #define ERR_DISPENSETEST_REMOTE_FAILED_CONN  -698
407 #define ERR_DISPENSETEST_REMOTE_SERVER_SHUT  -699
408 #define ERR_DISPENSETEST_REMOTE_ZMQ_MISSING  -700
409
410 // final test errors
411 #define ERR_FINALTEST_GENERAL_ERROR          -701
412 #define ERR_FINALTEST_INVALID_MODE           -702
```

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413 #define ERR_FINALTEST_CREATE_CSV_FILE        -703
414 #define ERR_FINALTEST_CREATE_IMAGE_FILE      -704
415 #define ERR_FINALTEST_TESTS_FAILED           -705
416
417 // prefocus errors
418 #define ERR_PREFOCUS_CREATE_IMAGE_FILE       -711
419 #define ERR_PREFOCUS_NOISE_TEST              -712
420 #define ERR_PREFOCUS_ADJ_LIMIT               -713
421
422 // laser line analysis errors
423 #define ERR_LASERLINE_MEM_ALLOC              -721
424 #define ERR_LASERLINE_NOT_LOCATED            -722
425
426 // gap/uv offset look up table errors
427 #define ERR_GAP_MEASURE_LUT_EXISTS           -725
428 #define ERR_GAP_MEASURE_LUT_DATA_FORMAT      -726
429 #define ERR_GAP_MEASURE_LUT_OUT_OF_RANGE     -727
430 #define ERR_GAP_MEASURE_CREATE_CSV_FILE      -728
431
432 // laser spot analysis errors
433 #define ERR_LASER_SPOT_NODATA_ACQUIRED       -730
434 #define ERR_LASER_SPOT_DATA_NOT_VALID        -731
435 #define ERR_LASER_SPOT_MEM_ALLOC             -732
436 #define ERR_LASER_SPOT_TIMEOUT               -733
437 #define ERR_LASER_SPOT_MISSING_CURFRAME      -734
438 #define ERR_LASER_SPOT_Z_OUT_RANGE           -735
439 #define ERR_LASER_SPOT_FWHM_FAILED           -736
440
441 // edge detection error
442 #define ERR_EDGE_CANNOT_RESOLVE_HOUGHLINES   -738
443 #define ERR_EDGE_ANGLE_OUTOF_TOLERANCE       -739
444 #define ERR_EDGE_NOT_ENOUGH_EDGEDATA         -740
445
446 // laser bars analysis errors
447 #define ERR_LASER_BAR_NO_PHASE_DATA_FOUND            -741
448 #define ERR_LASER_BAR_NO_PROFILE_IMAGE               -742
449 #define ERR_LASER_BAR_PROFILE_DATA_NOMEMORY          -743
450 #define ERR_LASER_BAR_CANNOT_LOG_DATA                -744
451 #define ERR_LASER_BAR_CANNOT_FOUND_LPM_AREA          -745
452 #define ERR_LASER_BAR_NO_DATA                        -746
453 #define ERR_LASER_BAR_NO_ROI_FOUND                   -747
454 #define ERR_LASER_BAR_NO_FEATURE_FOUND               -748
455 #define ERR_LASER_BAR_UNABLE_TO_LOCATE_LD            -749
456
457 // data line integrity test errors
458 #define ERR_DATALINETEST_CREATE_CSV_FILE          -751
459 #define ERR_DATALINETEST_CREATE_IMAGE_FILE        -752
460 #define ERR_DATALINETEST_LOAD_GOLDEN_FILE         -753
461 #define ERR_DATALINETEST_INCONSISTENT_IMAGE_FORMAT -754
462 #define ERR_DATALINETEST_TEST_FAILED              -755
463
464 // fixture calibration file errors
```

```
465 #define ERR_VALID_FIXTURE_AT_TABLE_POS_DUP   -801
466 #define ERR_VALID_FIXTURE_NOT_FOUND_POS1     -802
467 #define ERR_VALID_FIXTURE_NOT_FOUND_POS2     -803
468 #define ERR_VALID_FIXTURE_NOT_FOUND_POS3     -804
469 #define ERR_VALID_FIXTURE_NOT_FOUND_POS4     -805
470
471 // calibration and correction for vignetting and distortion
472 #define ERR_VISCAL_IMAGE_IS_INVALID                        -811
473 #define ERR_VISCAL_NO_CALIBRATION_PATTERN_DEFINED          -812
474 #define ERR_VISCAL_NO_CALIBRATION_PATTERN_FOUND            -813
475 #define ERR_VISCAL_FAILED_DO_DISTORTION_CALIBRATION        -814
476 #define ERR_VISCAL_NON_DISTORTION_CALIBRATION_DATA         -815
477 #define ERR_VISCAL_IMAGE_SIZE_DIFFERENT_THAN_CALIBRATION   -816
478 #define ERR_VISCAL_CAMERA_TRANSFORM_IS_DIFFERENT           -817
479 #define ERR_VISCAL_NOT_VALID_CAMERA_NUMBER                 -818
480 #define ERR_VISCAL_VIGNETTING_MEM_ALLOC                    -819
481 #define ERR_VISCAL_CANNOT_SAVE_CALIBRATION_DATA            -820
482 #define ERR_VISCAL_IMAGE_TOO_BRIGHT                        -821
483 #define ERR_VISCAL_IMAGE_TOO_DARK                          -822
484 #define ERR_VISCAL_NON_VIGNETTE_CALIBRATION_DATA           -823
485
486 // device data cache errors
487 #define ERR_DATA_CACHE_FILE_OPEN             -831
488 #define ERR_DATA_CACHE_FILE_READ             -832
489 #define ERR_DATA_CACHE_FILE_WRITE            -833
490 #define ERR_DATA_CACHE_TYPE                  -834
491 #define ERR_DATA_CACHE_VERSION               -835
492 #define ERR_DATA_CACHE_MAC_ADDRESS           -836
493 #define ERR_DATA_CACHE_SIZE                  -837
494 #define ERR_DATA_CACHE_WORK_ID_MISMATCH      -838
495 #define ERR_DATA_CACHE_NOT_FOUND             -839
496 #define ERR_DATA_CACHE_NOT_CREATED           -840
497
498 // configuration error second group
499 #define ERR_CONFIG_EXISTS_PIXIDSYS           -850
500 #define ERR_CONFIG_PARAM_TOO_MANY_CHARS      -851
501 #define ERR_CONFIG_PYTHON_NOT_SUPPORTED      -852
502 #define ERR_CONFIG_PYTHON_DUAL_FIXTURE       -853
503
504 // led alignment errors
505 #define ERR_LED_INVALID_NOMINAL_POSDATA      -860
506 #define ERR_LED_CANNOT_OPEN_FILE             -861
507 #define ERR_LED_FOUND_LESS_BLOBS             -862
508 #define ERR_LED_FILE_PATH_NULL               -863
509 #define ERR_LED_MISSING_APERTURE_SEARCH_DATA -864
510 #define ERR_LED_MISIING_TAB_SEARCH_DATA      -865
511 #define ERR_LED_RADIAL_ERROR                 -866
512
513 // Beam Profiler errors
514 #define ERR_BEAM_PROFILER_IMAGE_FORMAT       -880
515 #define ERR_BEAM_PROFILER_NUM_SPOT           -881
516 #define ERR_BEAM_PROFILER_INVALID_PARAM      -882
```

```
517 #define ERR_BEAM_PROFILER_NUM_PIXELS         -883
518
519 // Optimize errors
520 #define ERR_OPTIMIZE_INVALID_HANDLE          -890
521 #define ERR_OPTIMIZE_INVALID_ACTION          -891
522 #define ERR_OPTIMIZE_INVALID_FEATURE         -892
523 #define ERR_OPTIMIZE_INVALID_METHOD          -893
524
525 // process tracking (adaptive intelligence) errors
526 #define ERR_PROC_TRACK_FILE_OPEN             -901
527 #define ERR_PROC_TRACK_FILE_READ             -902
528 #define ERR_PROC_TRACK_FILE_WRITE            -903
529 #define ERR_PROC_TRACK_TYPE                  -904
530 #define ERR_PROC_TRACK_VERSION               -905
531 #define ERR_PROC_TRACK_MAC_ADDRESS           -906
532 #define ERR_PROC_TRACK_SIZE                  -907
533 #define ERR_PROC_TRACK_PLACE_X_ADJ_LIMIT     -908
534 #define ERR_PROC_TRACK_PLACE_Y_ADJ_LIMIT     -909
535 #define ERR_PROC_TRACK_PLACE_Z_ADJ_LIMIT     -910
536 #define ERR_PROC_TRACK_PLACE_TX_ADJ_LIMIT    -911
537 #define ERR_PROC_TRACK_PLACE_TY_ADJ_LIMIT    -912
538
539 // from KSwsc.exe
540 #define ERR_KSWSC_EXE_NOT_FOUND              -951
541 #define ERR_KSWSC_CREATE_PROCESS             -952
542 #define ERR_KSWSC_GET_EXIT_CODE              -953
543 #define ERR_KSWSC_WEB_SERVICE_TIMEOUT        -954
544 #define ERR_KSWSC_INVALID_ARG                -955
545 #define ERR_KSWSC_GETSN_WEB_SERVICE_EXCEPTION -956
546 #define ERR_KSWSC_GETSN_WEB_SERVICE_PARSE    -957
547 #define ERR_KSWSC_GETSN_WEB_SERVICE_FAILURE  -958
548 #define ERR_KSWSC_GETMODEL_WEB_SERVICE_EXCEPTION -959
549 #define ERR_KSWSC_GETMODEL_WEB_SERVICE_PARSE -960
550 #define ERR_KSWSC_GETMODEL_WEB_SERVICE_FAILURE -961
551 #define ERR_KSWSC_WRITE_RESULT_FILE          -962
552 #define ERR_KSWSC_READ_RESULT_FILE           -963
553 #define ERR_KSWSC_PARSE_RESULT_FILE          -964
554
555 // from KSbitmap.h which is part of KShexsight library
556 #define ERR_BITMAP_BAD_FILE_SPEC             -1001
557 #define ERR_BITMAP_OPEN_FILE_FAILED          -1002
558 #define ERR_BITAMP_LOAD_BAD_BPP              -1003
559 #define ERR_BITMAP_MEM_ALLOC                 -1004
560 #define ERR_BITMAP_NO_IMAGE                  -1005
561 #define ERR_BITMAP_NEW_BAD_IMAGE_PARAMS      -1006
562
563 // from KShexsight.h which is part of KShexsight library
564 #define ERR_HS_MFC_INIT_FAIL                 -1101
565 #define ERR_HS_NOT_INIT                      -1102
566 #define ERR_HS_CREATE_ACTIVEX_FAIL           -1103
567 #define ERR_HS_CONFIG_NOT_LOADED             -1104
568 #define ERR_HS_INVALID_MATCH_INDEX           -1105
```

```
569 #define ERR_HS_LOAD_CONFIG                  -1106
570 #define ERR_HS_LOAD_MODEL_DATABASE          -1107
571 #define ERR_HS_ADD_MODEL                    -1108
572 #define ERR_HS_SAVE_CONFIG                  -1109
573 #define ERR_HS_SAVE_MODEL_DATABASE          -1110
574
575 // from CIpc class which is defined in KSipc.h and used by both the KSvision
     library as well as the KSvisionProxy executable to manage device camera
     functionality
576 #define ERR_IPC_NOT_INIT                    -1201
577 #define ERR_IPC_CANT_LOCK                   -1202
578 #define ERR_IPC_CREATE_PROCESS              -1203
579 #define ERR_IPC_CANT_CREATE_MUTEX           -1204
580 #define ERR_IPC_CANT_CREATE_FILE_MAP        -1205
581 #define ERR_IPC_CANT_MAP_VIEW_OF_FILE       -1206
582 #define ERR_IPC_NOT_ALLOWED_THIS_OWNER      -1207
583 #define ERR_IPC_TIMEOUT                     -1208
584 #define ERR_IPC_CRITICAL_ERROR              -1209
585 #define ERR_IPC_ABORTED_FROM_TIMEOUT        -1210
586 #define ERR_IPC_ABORTED_FROM_CRITICAL_ERROR -1211
587
588 // from KSscanner.h which is part of the KSscanner library
589 #define ERR_SC_COINITIALIZE                 -1301
590 #define ERR_SC_CREATE_OBJECT                -1302
591 #define ERR_SC_CREATE_EVENT_SINK            -1303
592 #define ERR_SC_METHOD_OPEN                  -1304
593 #define ERR_SC_METHOD_CLOSE                 -1305
594 #define ERR_SC_METHOD_GETSCANNERS           -1306
595 #define ERR_SC_SNAPI_NO_SCANNERS_FOUND      -1307
596 #define ERR_SC_SNAPI_MULT_SCANNERS_FOUND    -1308
597 #define ERR_SC_METHOD_REGISTER_EVENTS       -1309
598 #define ERR_SC_METHOD_PULL_TRIGGER          -1310
599 #define ERR_SC_METHOD_RELEASE_TRIGGER       -1311
600 #define ERR_SC_SCAN_NO_LABEL_DETECTED       -1312
601 #define ERR_SC_PARSE_SCANNER_ID_ERROR       -1313
602 #define ERR_SC_PARSE_LABEL_ERROR            -1314
603 #define ERR_SC_COM_ERROR                    -1315
604
605 // KSvisionty.dll error codes - only applicable for configurations with ksprocess
      configured with CameraType=5 (a custom camera type)
606 #define ERR_TY_SUBI_FIND                    -1401
607 #define ERR_TY_SUBI_READCHIPID              -1402
608 #define ERR_TY_SUBI_PERFORDDEVICECLOSESEQ   -1403
609 #define ERR_TY_SUBI_PERFORDDEVICEOPENSEQ    -1404
610 #define ERR_TY_SUBI_GETDEVICELOCATION       -1405
611 #define ERR_TY_SUBI_IMAGEDATA               -1406
612 #define ERR_TY_SUBI_SET_OPMODE_SLEEP        -1407
613 #define ERR_TY_SUBI_SET_IPMODE_RAW          -1408
614 #define ERR_TY_SUBI_SET_OPMODE_RUN          -1419
615 #define ERR_TY_OPOG_CALDATAWRITE            -1410
616 #define ERR_TY_OPOG_INITIALIZATION          -1411
617 #define ERR_TY_OPOG_DEINITIALIZATION        -1412
```

```
618  #define ERR_TY_OPOG_PERFORMONEPOINTOFFSET    -1413
619  #define ERR_TY_OPOG_APPLYONEPOINTOFFSETANDGAIN -1414
620  #define ERR_TY_OPOG_SHUTTERPOLARITYCHECK     -1415
621  #define ERR_TY_OPOG_OPENSHUTTER              -1416
622  #define ERR_TY_OPOG_SHUTTERLEAKAGECHECK      -1417
623  #define ERR_TY_OPOG_THETAZASWEEP             -1418
624
625  // from KSkeyence.h which is part of the KSkeyence library
626  #define ERR_KL_FUNC_NAK_COMMAND              -1501
627  #define ERR_KL_FUNC_NAK_COMMAND_LENGTH       -1502
628  #define ERR_KL_FUNC_NAK_TIMEOUT              -1503
629  #define ERR_KL_FUNC_NAK_CHECKSUM             -1504
630  #define ERR_KL_FUNC_NAK_INVALID_STATE        -1505
631  #define ERR_KL_FUNC_NAK_OTHER                -1506
632  #define ERR_KL_FUNC_NAK_PARAMETER            -1507
633  #define ERR_KL_FUNC_NAK_OUT_STAGE            -1508
634  #define ERR_KL_FUNC_NAK_OUT_HEAD_NUM         -1509
635  #define ERR_KL_FUNC_NAK_OUT_INVALID_CALC     -1510
636  #define ERR_KL_FUNC_NAK_OUT_VOID             -1511
637  #define ERR_KL_FUNC_NAK_INVALID_CYCLE        -1512
638  #define ERR_KL_FUNC_NAK_CTRL_ERROR           -1513
639  #define ERR_KL_FUNC_NAK_SRAM_ERROR           -1514
640  #define ERR_KL_FUNC_ERR_OPEN_DEVICE          -1515
641  #define ERR_KL_FUNC_ERR_NO_DEVICE            -1516
642  #define ERR_KL_FUNC_ERR_SEND                 -1517
643  #define ERR_KL_FUNC_ERR_RECEIVE              -1518
644  #define ERR_KL_FUNC_ERR_TIMEOUT              -1519
645  #define ERR_KL_FUNC_ERR_NODATA               -1520
646  #define ERR_KL_FUNC_ERR_NOMEMORY             -1521
647  #define ERR_KL_FUNC_ERR_DISCONNECT           -1522
648  #define ERR_KL_FUNC_ERR_UNKNOWN              -1523
649  #define ERR_KL_FLOATRESULT_RANGEOVER_P       -1524
650  #define ERR_KL_FLOATRESULT_RANGEOVER_N       -1525
651  #define ERR_KL_FLOATRESULT_WAITING           -1526
652  #define ERR_KL_FLOATRESULT_ALARM             -1527
653  #define ERR_KL_FLOATRESULT_INVALID           -1528
654  #define ERR_KL_FLOATRESULT_UNKNOWN           -1529
655  #define ERR_KL_DLL_KEYENCE_NOT_FOUND         -1530
656  #define ERR_KL_DLL_KEYENCE_CANNOT_LOAD       -1531
657  #define ERR_KL_DLL_KEYENCE_GET_FUNC_ADDR     -1532
658
659  // from KSlightwave.h which is part of the KSlightwave library
660  #define ERR_LW_OPEN_PORT                     -1601
661  #define ERR_LW_SET_PORT_BUF_SIZE             -1602
662  #define ERR_LW_GET_COMM_DEFAULT_CONFIG       -1603
663  #define ERR_LW_SET_COMM_CONFIG               -1604
664  #define ERR_LW_GET_COMM_STATE                -1605
665  #define ERR_LW_SET_COMM_STATE                -1606
666  #define ERR_LW_GET_COMM_TIMEOUTS             -1607
667  #define ERR_LW_SET_COMM_TIMEOUTS             -1608
668  #define ERR_LW_PURGE_COMM_BUFFERS            -1609
669  #define ERR_LW_READ                          -1610
```

```
670 #define ERR_LW_WRITE                        -1611
671 #define ERR_LW_RESPONSE_TOO_LARGE           -1612
672 #define ERR_LW_RESPONSE_TIMEOUT             -1613
673 #define ERR_LW_INVALID_COMPORT_NUM          -1614
674
675 // from KSdatabase.h which is part of the KSdatabase library
676 // TODO_RLB - add support for these in the KS_GetError() funnction
677 #define ERR_DB_DBMS_BACKEND_CANNOT_BE_LOADED -1700
678 #define ERR_DB_CANNOT_CONNECT_TO_DATABASE    -1701
679 #define ERR_DB_SQL_COMMAND_FAILED            -1702
680 #define ERR_DB_NO_COLUMNS_IN_TABLE           -1703
681 #define ERR_DB_TABLE_NOT_DEFINED             -1704
682 #define ERR_DB_COLUMN_DEFINITION_INCOMPLETE  -1705
683 #define ERR_DB_COLUMN_NOT_DEFINED            -1706
684 #define ERR_DB_EMPTY_SELECT_OR_FROM_LIST     -1707
685 #define ERR_DB_CSV_INDEX_OUT_OF_RANGE        -1708
686 #define ERR_DB_COLUMN_INDEX_OUT_OF_RANGE     -1709
687 #define ERR_DB_QUERY_RESULT_COLUMN_COUNT     -1710
688 #define ERR_DB_UNHANDLED_EXCEPTION           -1711
689 #define ERR_DB_NO_TABLE_DEFINITIONS_IN_MAP   -1712
690
691 // from KSmettlertoledo.h which is part of the KSmettlertoledo library
692 #define ERR_MT_OPEN_PORT                     -1801
693 #define ERR_MT_SET_PORT_BUF_SIZE             -1802
694 #define ERR_MT_GET_COMM_DEFAULT_CONFIG       -1803
695 #define ERR_MT_SET_COMM_CONFIG               -1804
696 #define ERR_MT_GET_COMM_STATE                -1805
697 #define ERR_MT_SET_COMM_STATE                -1806
698 #define ERR_MT_GET_COMM_TIMEOUTS             -1807
699 #define ERR_MT_SET_COMM_TIMEOUTS             -1808
700 #define ERR_MT_PURGE_COMM_BUFFERS            -1809
701 #define ERR_MT_READ                          -1810
702 #define ERR_MT_WRITE                         -1811
703 #define ERR_MT_RESPONSE_TOO_LARGE            -1812
704 #define ERR_MT_RESPONSE_TIMEOUT              -1813
705
706 // from KSscannercx.h which is part of the KSscannercx library
707 #define ERR_CX_OPEN_PORT                     -1901
708 #define ERR_CX_SET_PORT_BUF_SIZE             -1902
709 #define ERR_CX_GET_COMM_DEFAULT_CONFIG       -1903
710 #define ERR_CX_SET_COMM_CONFIG               -1904
711 #define ERR_CX_GET_COMM_STATE                -1905
712 #define ERR_CX_SET_COMM_STATE                -1906
713 #define ERR_CX_GET_COMM_TIMEOUTS             -1907
714 #define ERR_CX_SET_COMM_TIMEOUTS             -1908
715 #define ERR_CX_PURGE_COMM_BUFFERS            -1909
716 #define ERR_CX_READ                          -1910
717 #define ERR_CX_WRITE                         -1911
718 #define ERR_CX_RESPONSE_TOO_LARGE            -1912
719 #define ERR_CX_RESPONSE_TIMEOUT              -1913
720 #define ERR_CX_INVALID_COMMPORT_NUM          -1914
721 #define ERR_CX_INVALID_FORMAT                -1915
```

```
722  #define ERR_CX_MISSING_LOCATION1             -1920
723  #define ERR_CX_MISSING_LOCATION2             -1921
724  #define ERR_CX_MISSING_LOCATION3             -1922
725  #define ERR_CX_MISSING_LOCATION4             -1923
726  #define ERR_CX_MISSING_LOCATION5             -1924
727  #define ERR_CX_MISSING_LOCATION6             -1925
728  #define ERR_CX_MISSING_MULTI_LOCATION        -1930
729  #define ERR_CX_MISSING_ONE_LOCATION          -1931
730  #define ERR_CX_MISSING_TWO_LOCATION          -1932
731  #define ERR_CX_MISSING_THREE_LOCATION        -1933
732  #define ERR_CX_MISSING_FOUR_LOCATION         -1934
733  #define ERR_CX_SORT_ERROR                    -1940
734  #define ERR_CX_MAX_COMMPORT_REACHED          -1941
735
736  // from KSdataray.h which is part of the KSdataray laser profiler library
737  // TODO: This whold section is missing from ErrorExtras in Pixid.
738  #define ERR_DT_NOT_INITIALIZED               -2001
739  #define ERR_DT_DRIVER_START_FAILED           -2002
740  #define ERR_DT_DEVICE_START_FAILED           -2003
741  #define ERR_DT_DEVICE_STOP_FAILED            -2004
742  #define ERR_DT_DATA_NOT_READY                -2005
743  #define ERR_DT_DEVICE_NOT_PRESENT            -2006
744  #define ERR_DT_GRAB_TIMEOUT                  -2007
745  #define ERR_DT_NO_MEMORY                     -2008
746  #define ERR_DT_MODE_UNSUPPORT                -2009
747  #define ERR_DT_BAD_CAM_NUM                   -2010
748
749  // from KSaardvark.h which is part of the KSaardvark library
750  #define ERR_AA_OPEN_PORT                     -2101
751  #define ERR_AA_CLOSE_PORT                    -2102
752  #define ERR_AA_CONFIG                        -2103
753  #define ERR_AA_INCOMPATIBLE_DEVICE           -2104
754  #define ERR_AA_REG_GET_WRITE_ADR             -2105
755  #define ERR_AA_REG_GET_READ_VAL              -2106
756  #define ERR_AA_REG_SET                       -2107
757  #define ERR_AA_REG_SET_ARG_INVALID           -2108
758  #define ERR_AA_REG_SET_NOT_INITIALIZED       -2109
759  #define ERR_AA_REG_GET_ARG_INVALID           -2110
760  #define ERR_AA_REG_GET_NOT_INITIALIZED       -2111
761  #define ERR_AA_REG_VERIFY                    -2112
762  #define ERR_AA_NOT_ALL_DEVICES_FOUND         -2113
763  #define ERR_AA_INIT_LOAD_LIB                 -2114
764
765  // errors for KSvisionaa
766  #define ERR_KSVAA_UNABLE_TO_LOAD_LIBRARY     -2130
767  #define ERR_KSVAA_UNABLE_TO_LOAD_DRIVER      -2131
768  #define ERR_KSVAA_NO_DEVICES                 -2132
769  #define ERR_KSVAA_TOO_MANY_DEVICES           -2133
770  #define ERR_KSVAA_PORT_IN_USE                -2134
771  #define ERR_KSVAA_DEVICE_SERIAL_NUM_NF       -2135
772  #define ERR_KSVAA_UNABLE_TO_OPEN_DEVICE      -2136
773  #define ERR_KSVAA_INCOMPATIBLE_DEVICE        -2137
```

```
774 #define ERR_KSVAA_DEVICE_NOT_OPEN            -2138
775 #define ERR_KSVAA_DEVICE_NOT_INITIALIZED     -2139
776 #define ERR_KSVAA_WRITE_REGISTER_FAILED      -2140
777 #define ERR_KSVAA_BAD_DEVICE_NUM             -2141
778 #define ERR_KSVAA_SET_REG_ARG_INVALID        -2142
779 #define ERR_KSVAA_SET_REG_ADDR_INVALID       -2143
780 #define ERR_KSVAA_SET_REG_DATA_INVALID       -2144
781 #define ERR_KSVAA_READ_REGISTER_FAILED       -2145
782
783 // from KSgscontrol.h which is part of the KSgscontrol library - not required in ↵
     PIXID text tables
784 #define ERR_GS_OPEN_PORT                     -2201
785 #define ERR_GS_SET_PORT_BUF_SIZE             -2202
786 #define ERR_GS_GET_COMM_DEFAULT_CONFIG       -2203
787 #define ERR_GS_SET_COMM_CONFIG               -2204
788 #define ERR_GS_GET_COMM_STATE                -2205
789 #define ERR_GS_SET_COMM_STATE                -2206
790 #define ERR_GS_GET_COMM_TIMEOUTS             -2207
791 #define ERR_GS_SET_COMM_TIMEOUTS             -2208
792 #define ERR_GS_PURGE_COMM_BUFFERS            -2209
793 #define ERR_GS_READ                          -2210
794 #define ERR_GS_WRITE                         -2211
795 #define ERR_GS_RESPONSE_TOO_LARGE            -2212
796 #define ERR_GS_RESPONSE_TIMEOUT              -2213
797 #define ERR_GS_INVALID_CMD                   -2214
798 #define ERR_GS_INVALID_VALUE                 -2215
799 #define ERR_GS_UNEXPECTED_RESPONSE           -2216
800 #define ERR_GS_RESET_NOT_DETECTED            -2217
801 #define ERR_GS_I2C_FAILURE                   -2218
802 #define ERR_GS_I2C_WRITE_REGISTER            -2219
803 #define ERR_GS_I2C_VERIFY                    -2220
804 #define ERR_GS_USER_INPUT_ERROR              -2221
805
806 // from KSbnc.h which is part of the KSbnc library
807 #define ERR_BNC_OPEN_PORT                    -2301
808 #define ERR_BNC_SET_PORT_BUF_SIZE            -2302
809 #define ERR_BNC_GET_COMM_DEFAULT_CONFIG      -2303
810 #define ERR_BNC_SET_COMM_CONFIG              -2304
811 #define ERR_BNC_GET_COMM_STATE               -2305
812 #define ERR_BNC_SET_COMM_STATE               -2306
813 #define ERR_BNC_GET_COMM_TIMEOUTS            -2307
814 #define ERR_BNC_SET_COMM_TIMEOUTS            -2308
815 #define ERR_BNC_PURGE_COMM_BUFFERS           -2309
816 #define ERR_BNC_READ                         -2310
817 #define ERR_BNC_WRITE                        -2311
818 #define ERR_BNC_RESPONSE_TOO_LARGE           -2312
819 #define ERR_BNC_RESPONSE_TIMEOUT             -2313
820 #define ERR_BNC_COMMAND_UNKNOWN_ERROR        -2314
821 #define ERR_BNC_COMMAND_PREFIX_ERROR         -2315
822 #define ERR_BNC_COMMAND_MISSING_KEYWORD      -2316
823 #define ERR_BNC_COMMAND_INVALID_KEYWORD      -2317
824 #define ERR_BNC_COMMAND_MISSING_PARAM        -2318
```

```
825 #define ERR_BNC_COMMAND_INVALID_PARAM        -2319
826 #define ERR_BNC_COMMAND_QUERY_FORMAT_ERROR   -2320
827 #define ERR_BNC_COMMAND_INVALID_QUERY        -2321
828 #define ERR_BNC_COMMAND_INVALID              -2322
829 #define ERR_BNC_COMMAND_PARAM_OUT_OF_BOUND   -2323
830
831 // for KSlcosGE
832 #define ERR_LCOS_AARDVARK_LIB_LOAD           -2401
833 #define ERR_LCOS_AARDVARK_LIB_FNC_PTR        -2402
834 #define ERR_LCOS_AARDVARK_NOT_INIT           -2403
835 #define ERR_LCOS_TEMP_READING_FAIL           -2404
836 #define ERR_LCOS_OVERTEMP                    -2405
837
838 // for TARS MES data file generation
839 #define ERR_TARS_INI_FILE_NF                 -2501
840 #define ERR_TARS_FILE_PATH_PARAM_NF          -2502
841 #define ERR_TARS_MACHINE_ID_PARAM_NF         -2503
842 #define ERR_TARS_PROCESS_NAME_PARAM_NF       -2504
843 #define ERR_TARS_PROJECTION_SECT_PARAM_NF    -2505
844 #define ERR_TARS_SPECTRE_SECTION_PARAM_NF    -2506
845 #define ERR_TARS_LCOS_SECTION_PARAM_NF       -2507
846 #define ERR_TARS_GS_SECTION_PARAM_NF         -2508
847 #define ERR_TARS_ADHESIVE_SECT_PARAM_NF      -2509
848 #define ERR_TARS_PROJECTOR_MASK_PARAM_NF     -2510
849 #define ERR_TARS_SPECTRE_MASK_PARAM_NF       -2511
850 #define ERR_TARS_LCOS_MASK_PARAM_NF          -2512
851 #define ERR_TARS_GS_MASK_PARAM_NF            -2513
852 #define ERR_TARS_UNKNOWN_DATA_TAG_PARAM_NF   -2514
853 #define ERR_TARS_FAILED_TO_OPEN_DATA_FILE    -2515
854 #define ERR_TARS_INVALID_SERIAL_NUMBER       -2516
855 #define ERR_TARS_PARAM_LABEL_MAX_LENGTH      -2517
856 #define ERR_TARS_FILE_BACKUP_PATH_PARAM_NF   -2518
857 #define ERR_TARS_RETEST_FILE_PATH_PARAM_NF   -2519
858 #define ERR_TARS_RUN_TIS_PARAM_NF            -2520
859 #define ERR_TARS_TIS_SERIAL_NUM_NF           -2521
860 #define ERR_TARS_TIS_NETWORK_FAILURE         -2522
861 #define ERR_TARS_TIS_EXECUTE_FAILED          -2523
862 #define ERR_TARS_TIS_FAILED_TO_GET_EXIT_CODE -2524
863 #define ERR_TARS_TIS_UNKNOWN                 -2525
864 #define ERR_TARS_TIS_TISEXE_NF               -2526
865 #define ERR_TARS_TIS_TIMELIMIT_PARAM_NF      -2527
866 #define ERR_TARS_TIS_TIMED_OUT               -2528
867 #define ERR_TARS_TIS_CUSTOMER_NF             -2529
868 #define ERR_TARS_TIS_DIVISION_NF             -2530
869 #define ERR_TARS_PARAM_NF                    -2531
870
871 // for TRS MES data file generation
872 #define ERR_TRS_LOG_FILE_OPEN                -2540
873 #define ERR_TRS_LOG_DIR_EXISTS               -2541
874 #define ERR_TRS_INVALID_RECORD_DATA          -2542
875 #define ERR_TRS_FILE_PATH_PARAM_NF           -2543
876
```

```
877 // for Tacoma ASIC I2C communications
878 #define ERR_TAC_ASIC_RTN_NAK                -2545
879 #define ERR_TAC_ASIC_RTN_CHECKSUM_ERR       -2546
880 #define ERR_TAC_BLEMISH_DATA_VALIDATION     -2547
881
882 // for external alignment errors
883 #define ERR_ALIGN_EXTERNAL_FAILED_CONNECT   -2550
884 #define ERR_ALIGN_EXTERNAL_DISCONNECT       -2551
885 #define ERR_ALIGN_EXTERNAL_FAILED           -2552
886 #define ERR_ALIGN_EXTERNAL_MANUAL_DISCONNECT -2553
887 #define ERR_ALIGN_EXTERNAL_UNSAFE_INDEX     -2554
888 #define ERR_ALIGN_EXTERNAL_ZMQ_MISSING      -2555
889
890 // for Tacoma ASIC errors
891
892 // for GEEOLTest - gaps are to allow the error codes to align with
     KSvisionEOLTest project error codes
893 #define ERR_EOL_LCOS_CORNERS_NOT_FOUND      -5000
894 #define ERR_EOL_LCOS_CORNERS_OUT_OF_RANGE   -5001
895 #define ERR_EOL_UNIFORMITY_MEM_ALLOC        -5002
896 #define ERR_EOL_ANSI_MEM_ALLOC              -5003
897 #define ERR_EOL_LCOS_IMAGE_FILE_NOT_DEFINED -5004
898 #define ERR_EOL_IMAGE_SIZE_ERROR            -5005
899 #define ERR_EOL_ONE_OR_MORE_TESTS_FAILED    -5011
900 #define ERR_EOL_ROI_DATA_NOT_INITIALIZED    -5020
901 #define ERR_EOL_GS_CONTROL_LIB_LOAD         -5021
902 #define ERR_EOL_GS_CONTROL_LIB_FNC_PTR      -5022
903 #define ERR_EOL_GS_CONTROL_NOT_INIT         -5023
904 #define ERR_EOL_LED_THERMISTOR_FAIL         -5024
905 #define ERR_EOL_LED_THERMISTOR_READING_HIGH -5025
906 #define ERR_EOL_INADEQUATE_BIN_LEVEL        -5026
907 #define ERR_EOL_NOT_ALL_TESTS_COMPLETED     -5027
908 #define ERR_EOL_IMAGE_GREYSCALE_TOO_HIGH    -5028
909 #define ERR_EOL_INVALID_MTF_EXCEPTION_VALUE -5029
910 #define ERR_EOL_IMAGE_GREYSCALE_TOO_LOW     -5031
911 #define ERR_EOL_GREYSCALE_CHECK_ANSI        -5032
912 #define ERR_EOL_GREYSCALE_CHECK_ANSI_HDR    -5033
913 #define ERR_EOL_GREYSCALE_CHECK_ANSI_UNIFORMITY -5034
914 #define ERR_EOL_GREYSCALE_CHECK_MTF_HORZ_VERT -5035
915 #define ERR_EOL_ASPECT_RATIO                 -5036
916
917 #define ERR_EOL_MTF_SATURATED               -5062
918 #define ERR_EOL_MTF_CLIPPED                 -5063
919 #define ERR_EOL_UNIFORMITY_SATURATED        -5064
920 #define ERR_EOL_MTF_ROI_OUTSIDE_FOV         -5065
921
922 #define ERR_EOL_VIGNETTING_EXIST            -5070
923 #define ERR_EOL_VIGNETTING_IMAGE_FORMAT     -5071
924 #define ERR_EOL_VIGNETTING_IMAGE_VALUE_ZERO -5072
925 #define ERR_EOL_VIGNETTING_IMAGE_SIZE       -5073
926 #define ERR_EOL_VIGNETTING_DATA_SIZE        -5074
927 #define ERR_EOL_BLEMISH_CONV_EXCEPTION      -5075
```

```
928  #define ERR_EOL_INVALID_TEST_TYPE_VALUE      -5081
929  #define ERR_EOL_INVALID_TEST_STATE_VALUE     -5082
930  #define ERR_EOL_UNABLE_TO_WARMUP             -5083
931  #define ERR_EOL_UNABLE_TO_READ_CURRENT       -5084
932
```